M Answer-DSP Quiz # 2- Summer 26- All Laws
Date: 02 / 08 / 2026 & Time: 20 m (3 Marks)

MA- Q # 1:

Determine the output of an electrical system of impulse response function {0, 0.8, 0.99, 0.9, 1} when the input {0, 2.5, 5.0, 0} volts are applied

Solution:

h(n) = {0, 0.8, 0.99, 0.9, 1}……. Channel impulse response
x(n) = {0, 2.5, 5, 0} ……. Input data sequence

The direct convolution is given by: $x(n) \otimes h(n) = \sum_{m=0}^{n} h(m)x(n-m)$.

Lag n	h(m)	0	0	0	0	0.8	0.99	0.9	1	$x(n) \otimes h(n)$
0	x(n-m)	0	5	2.5	0	0	0	0	0	0
1		0	0	5	2.5	0	0	0	0	0
2		0	0	0	5	2.5	0	0	0	2
3		0	0	0	0	5	2.5	0	0	6.475
4		0	0	0	0	0	5	2.5	0	7.2
5		0	0	0	0	0	0	5	2.5	7
6		0	0	0	0	0	0	0	5	5
7		0	0	0	0	0	0	0	0	0

y(n) = {0, 0, 2, 6.475, 7.2, 7, 5, 0}

MA- Q # 2:

Obtain the cross-correlation $r_{12}(-\tau)$ between the waveforms $\upsilon_1(t)$ and $\upsilon_2(t)$ of Fig. 1

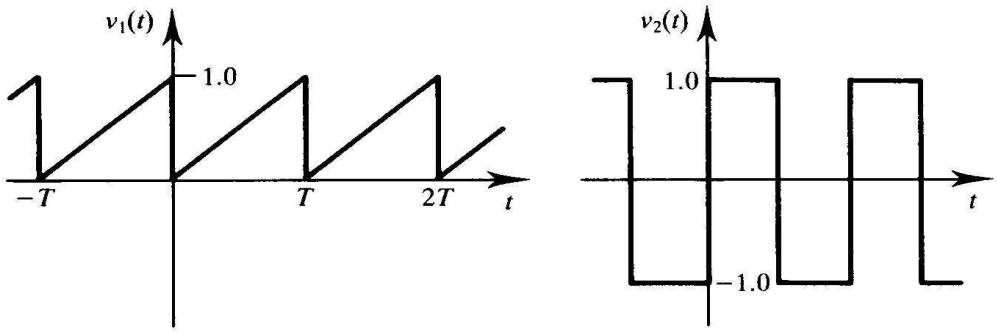


Fig. 1 The waveforms $\upsilon_1(t)$ and $\upsilon_2(t)$.

Solution:

Example 3.3

Obtain the cross-correlation $r_{12}(-\tau)$ between the waveforms $x_1(t)$ and $x_2(t)$ of Fig. 1

Solution

It is easy to express the waveforms analytically by dividing them into straight-line sections. It is only necessary to do this over one period, T, of the waveforms because $r_{12}(-\tau)$ will be periodic in τ with period T. Therefore, for $0 \le t \le T$, $\upsilon_1(t) = t/T$, and for $0 \le t \le T/2$, $\upsilon_2(t) = 1.0$, while for $T/2 \le t \le T$, $\upsilon_2(t) = -1.0$.

- The requirement is to obtain an expression for $r_{12}(-\tau)$, that is $\upsilon_2(t)$, the rectangular waveform, is to be shifted right w.r.t. $\upsilon_1(t)$.
- **For** $0 \le \tau \le T/2$, the situation is described by Fig. 3.7 which shows that $\upsilon_1(t)$ has to be multiplied by three consecutive section of $\upsilon_2(t)$ in which $\upsilon_2(t)$ has the consecutive value – 1, 1, -1.
- **For** $T/2 \le \tau \le T$, Fig. 3.8 applies in which the consecutive values of the set of $\upsilon_2(t)$ have changed to 1, -1, +1. This means there are two parts to the solution which must match at $\tau = T/2$.

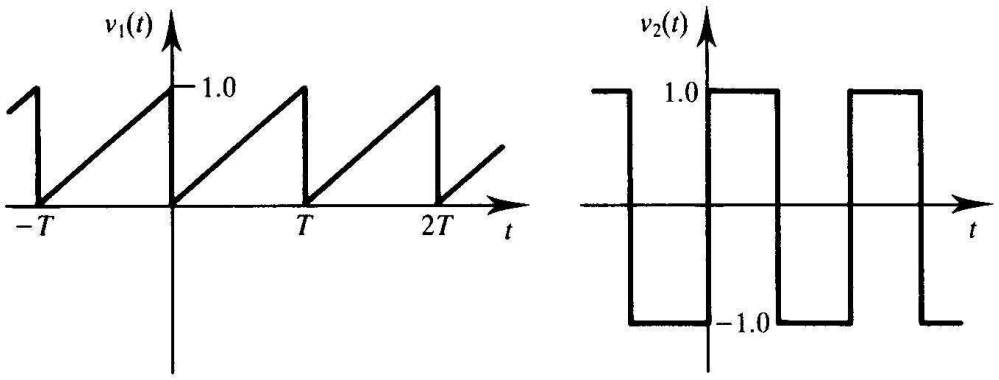


Fig. 3.6 The waveforms $\upsilon_1(t)$ and $\upsilon_2(t)$ for cross-correlation example.

Referring to Fig. 3.7, the cross-correlation is split into the three sections with boundaries at t = τ, t = τ+T/2, and t = T. Hence

$$r_{12}(-\tau) = \frac{1}{T}\int_0^T \upsilon_1(t)\upsilon_2(t-\tau)\,dt$$

$$= \frac{1}{T}\int_0^{\tau} \frac{t}{T}(-1)\,dt + \frac{1}{T}\int_{\tau}^{\tau+T/2} \frac{t}{T}(1)dt + \frac{1}{T}\int_{\tau+T/2}^{T} \frac{t}{T}(-1)dt$$

$$= \frac{-1}{T^2}\left[\frac{t^2}{2}\right]_0^{\tau} + \frac{1}{T^2}\left[\frac{t^2}{2}\right]_{\tau}^{\tau+T/2} - \frac{1}{T^2}\left[\frac{t^2}{2}\right]_{\tau+T/2}^{T}$$

$$r_{12}(-\tau) = -\frac{1}{4} + \frac{\tau}{T} \text{ for } 0 \le \tau \le \frac{T}{2} \qquad (3.11)$$

For $T/2 \le \tau \le T$, and referring to Fig. 3.8, it is seen that

$$r_{12}(-\tau) = \frac{1}{T}\int_0^{\tau-T/2} \frac{t}{T}(1)dt + \frac{1}{T}\int_{\tau-T/2}^{\tau} \frac{t}{T}(-1)dt + \frac{1}{T}\int_{\tau}^{T} \frac{t}{T}(1)dt$$

$$r_{12}(-\tau) = \frac{3}{4} - \frac{\tau}{T} \text{ for } \frac{T}{2} \le \tau \le T \qquad (3.12)$$

Substituting τ = T/2 into Equations 3.11 and 3.12 gives $r_{12}(-\tau)$ = 1/4 in both cases, confirming that the two functions match correctly. Figure 3.9 shows a plot of $r_{12}(-\tau)$ versus τ for $0 \leq \tau \leq T$.

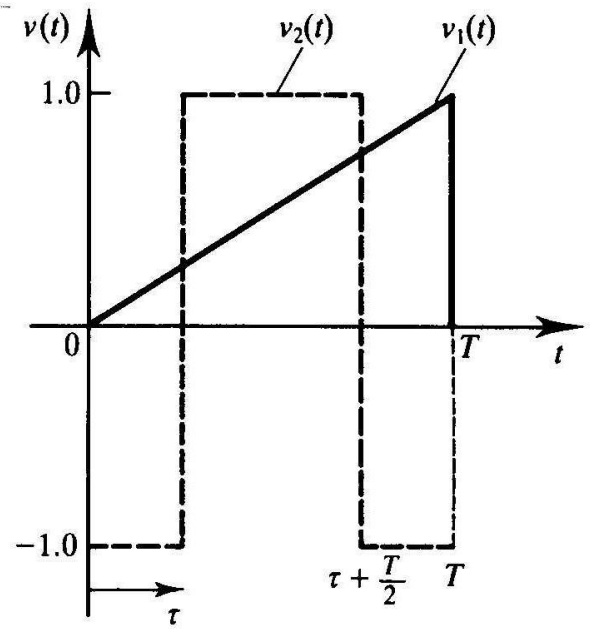


Fig. 3.7 Section of $\upsilon_2(t)$ for $0 \leq \tau \leq T$

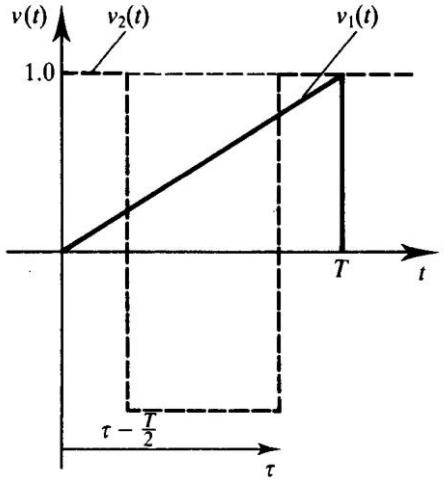


Fig. 3.8 Sections of $\upsilon_2(t)$ for $T/2 \leq \tau \leq T$.

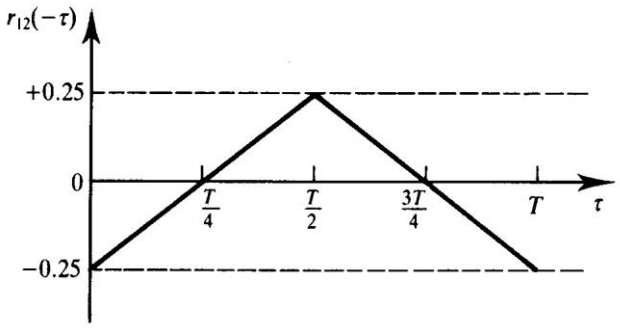


Fig. 3.9 $r_{12}(-\tau)$ as a function of τ.